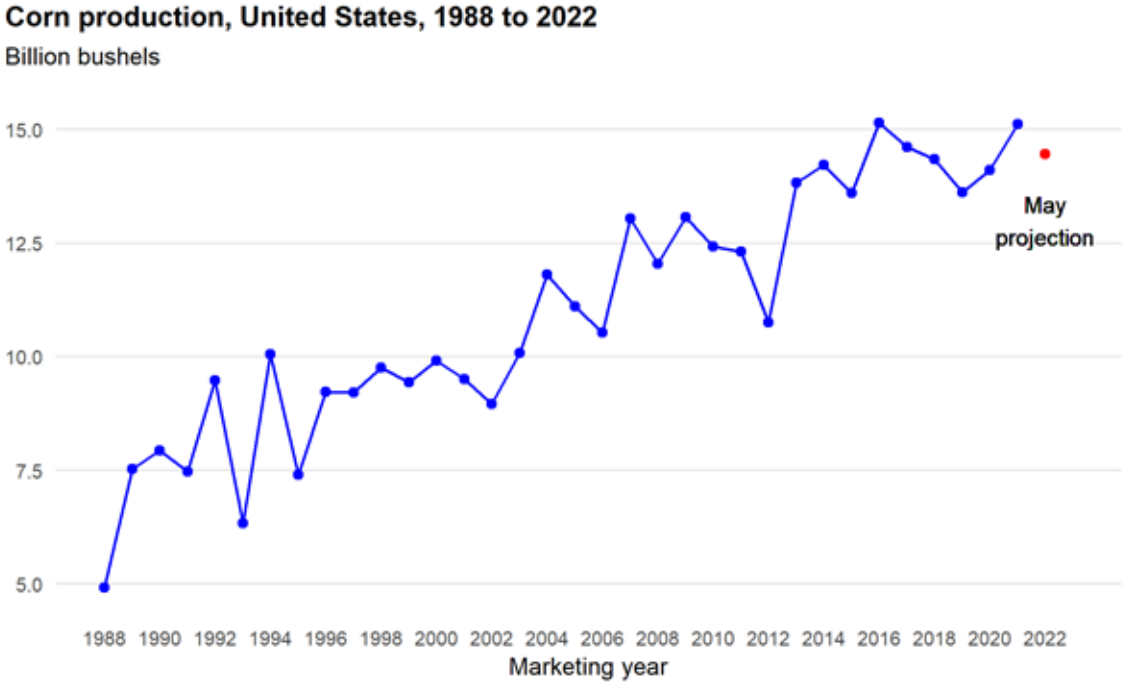
How do weather patterns affect price?

Data Bandits

Marco Conti Patrick Poggi Federico Mansutti Filippo Corna Marcin Kowalik

CORN IS FUNDAMENTAL IN THE US

Corn is the U.S.'s most produced, consumed, and exported crop.



Source: USDA, National Agricultural Statistics Service.

DATASETS

USDA NASS (Corn)

Includes data at a state level including yield, area harvested, and production.

Weather dataset from Kaggle (US)

Includes data that has the maximum and minimum temperature, and the precipitation levels from 1992-2021 spanning over 25000 stations in the US

USDA NASS (Corn Price)

Includes data that has the monthly price of corn

PRODUCTION

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Cleaning

Filtered data by:

- Removing empty values
- Formating data
- Computing Averages

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Results

We were able to find:

- Top producers
- Production in BU
  - 56 lbs.
- Price per BU
- Area harvested (ac)
- Filtered by states with over 1,000,000 acres

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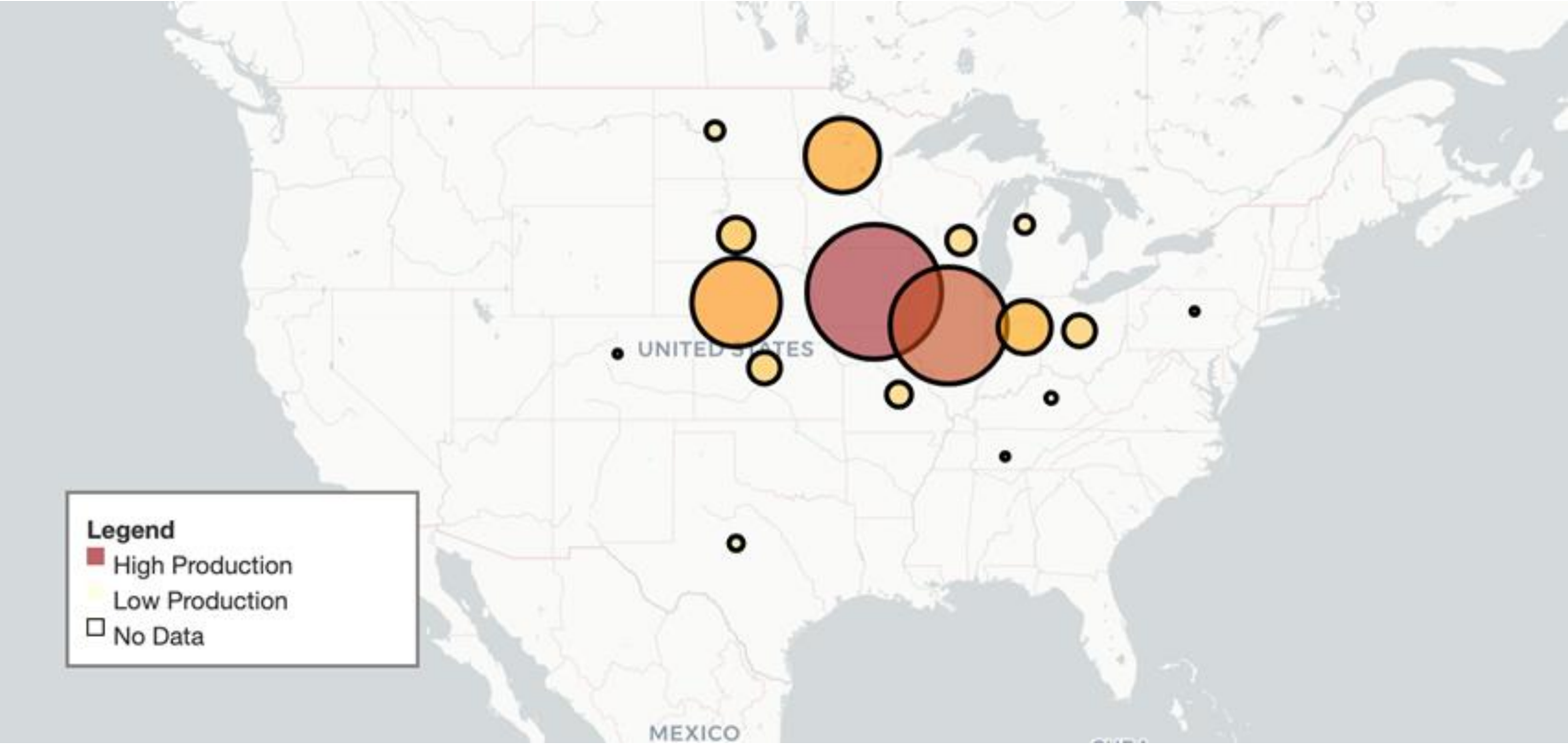
Use

With our data:

- Focus on the weather in this region
- Created a visual showing the top producers their average production

Production Visual

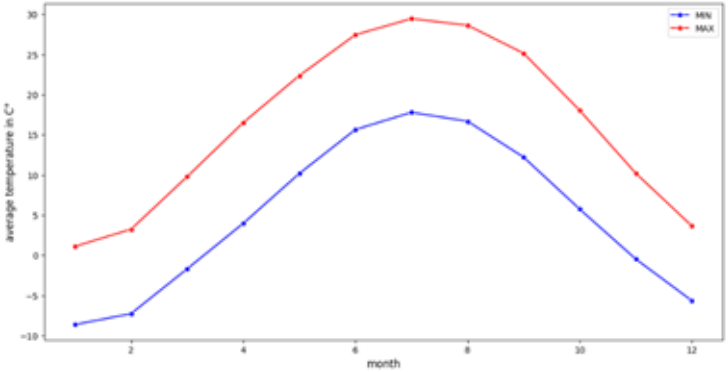


WEATHER DATA

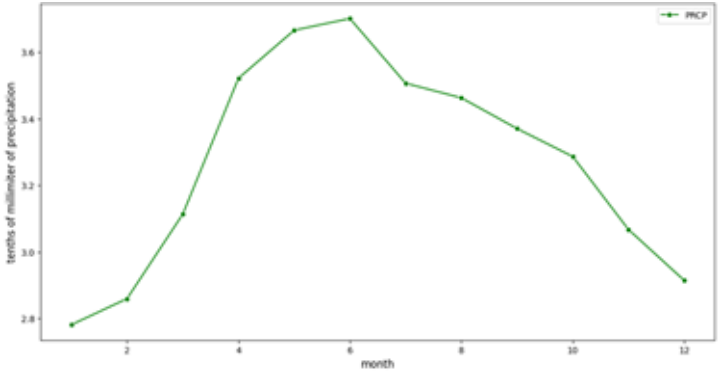
After cleaning some corrupted values

DATA FROM CORN BELT

AVERAGE TEMPERATURE

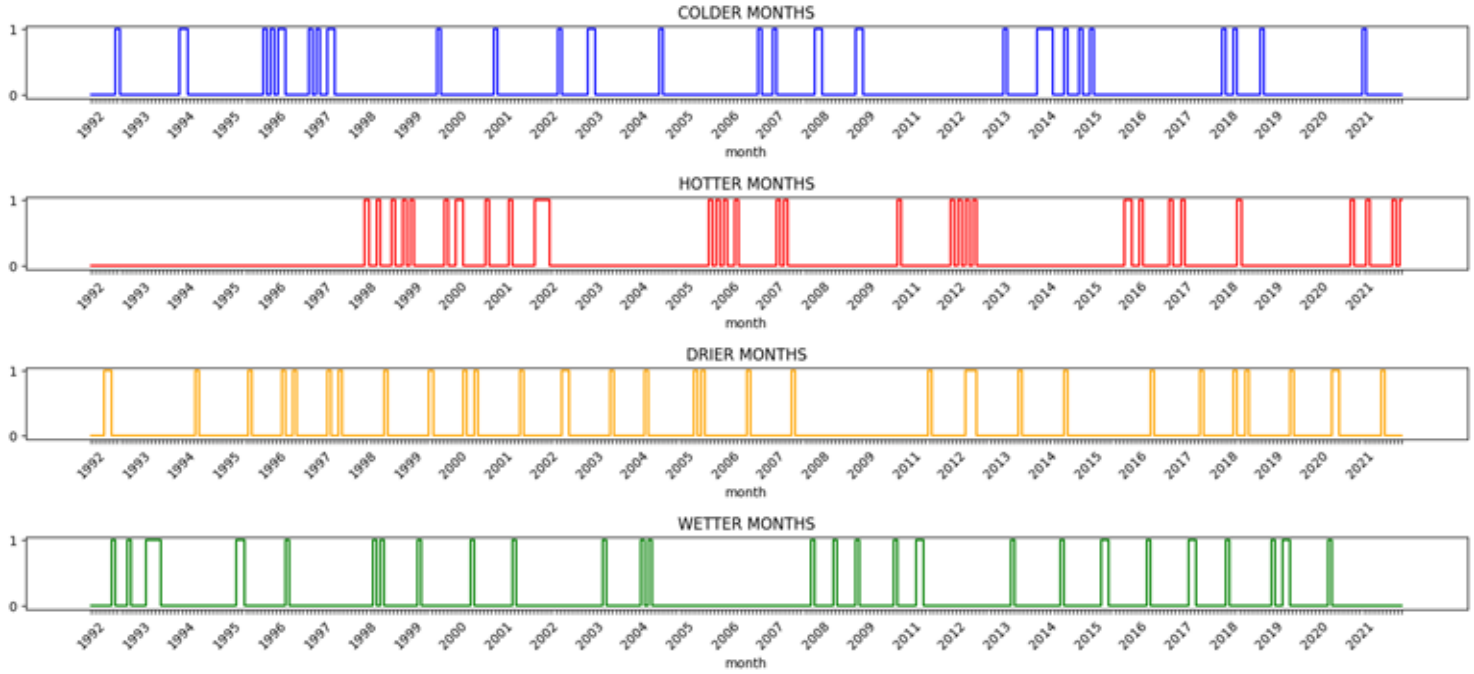


AVERAGE PRECIPITATIONS



WEATHER OUTLIERS

Months with more **outliers** = days with temperature or precipitations very different from to the monthly average



PRICE FLUCTUATION

A future is a financial contract obligating two parties to buy or sell an asset at a predetermined price on a specific future date.

Long-term futures: Contracts with a distant expiration

Short-term futures: Contracts with a near-term expiration

PRICE FLUCTUATION

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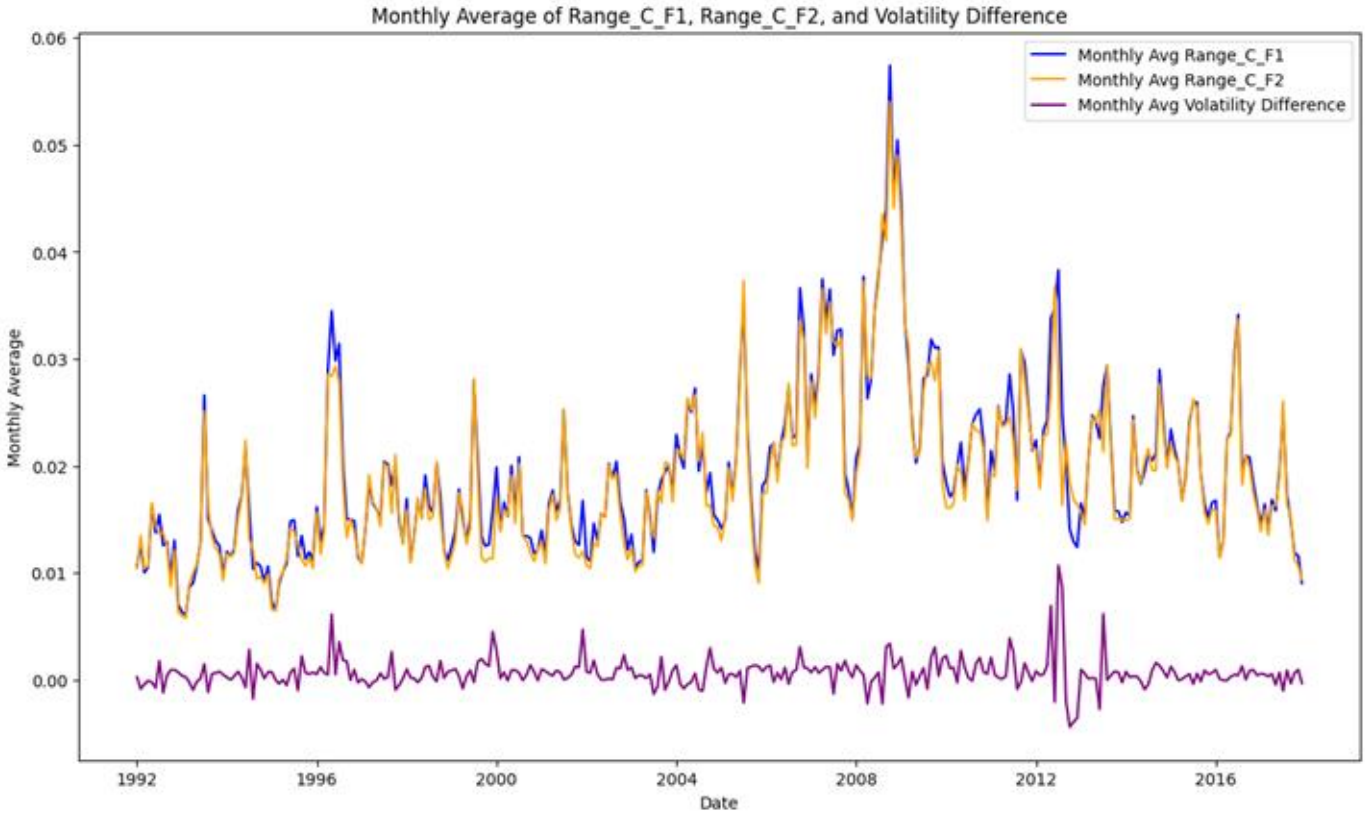
−

Short-term futures: Contracts with a near-term expiration

=

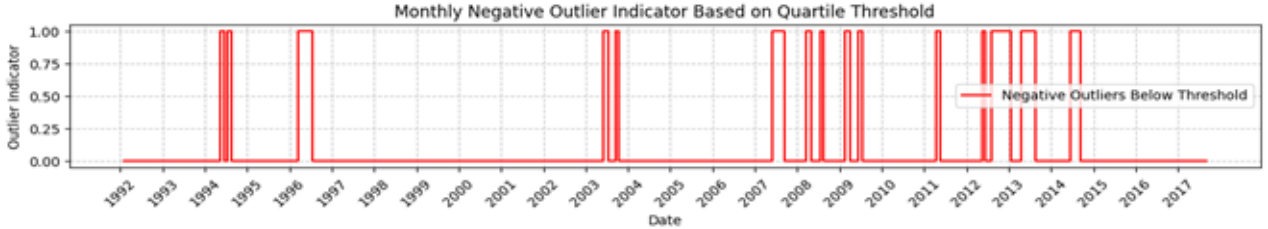
Difference: It reflects market volatility and expectations

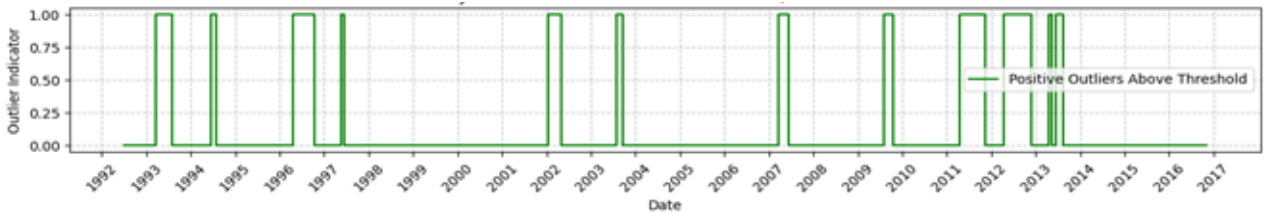
PRICE FLUCTUATION

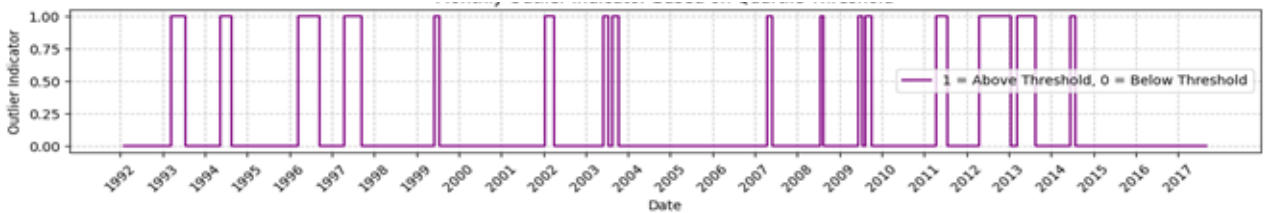


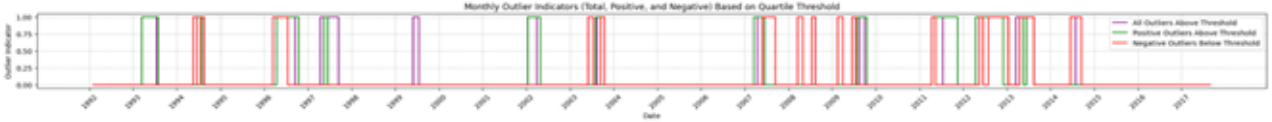
PRICE FLUCTUATION OUTLIERS

OUTLIERS: month with fluctuation outliers ("high number of positive or negative fluctuation of the price")









MONTHLY CORN PRICE

1

Gathered data from USDA website

First of all, i fetched data from USDA website though the available querying interface

MONTHLY CORN PRICE

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Data Cleaning and Wrangling

Filter out missing data and focus on major corn producers

MONTHLY CORN PRICE

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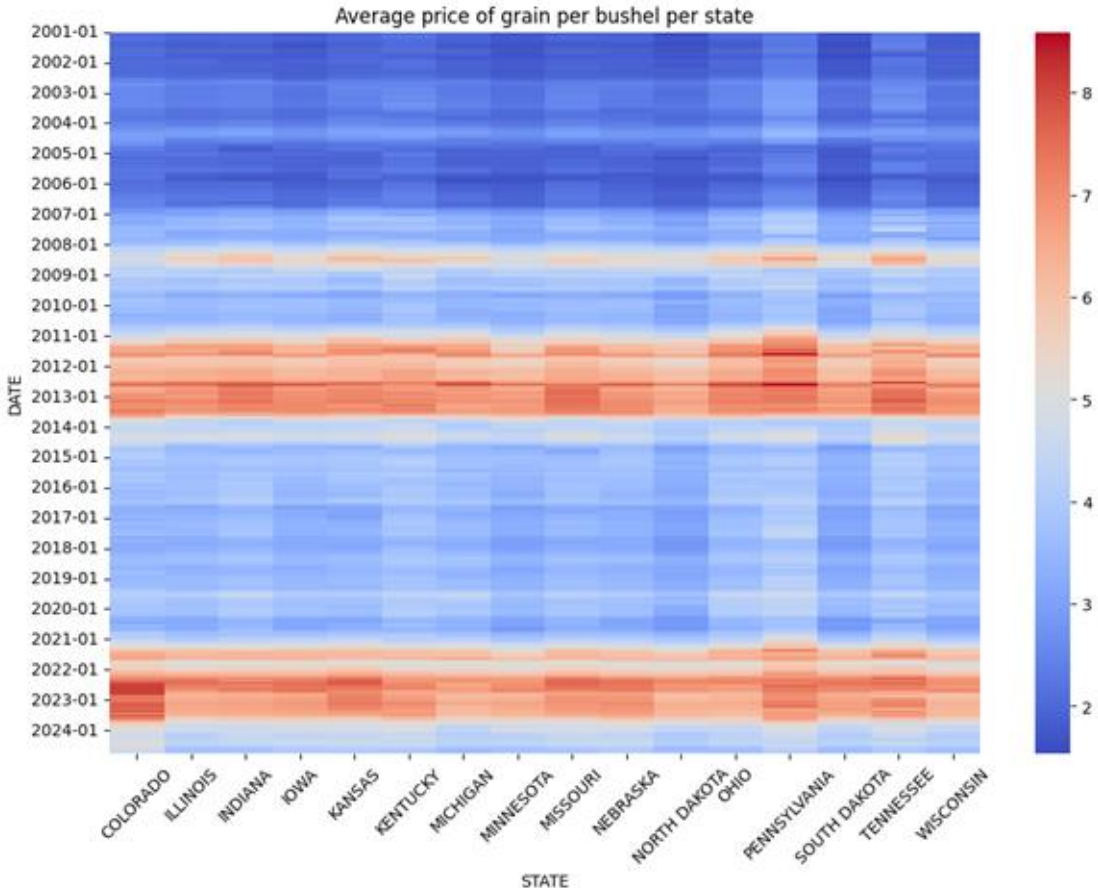
Exploratory Data Analysis

Investigate possible patterns or irregularities in the data to guide future Hypothesis Testing and project goals

MONTHLY CORN PRICE



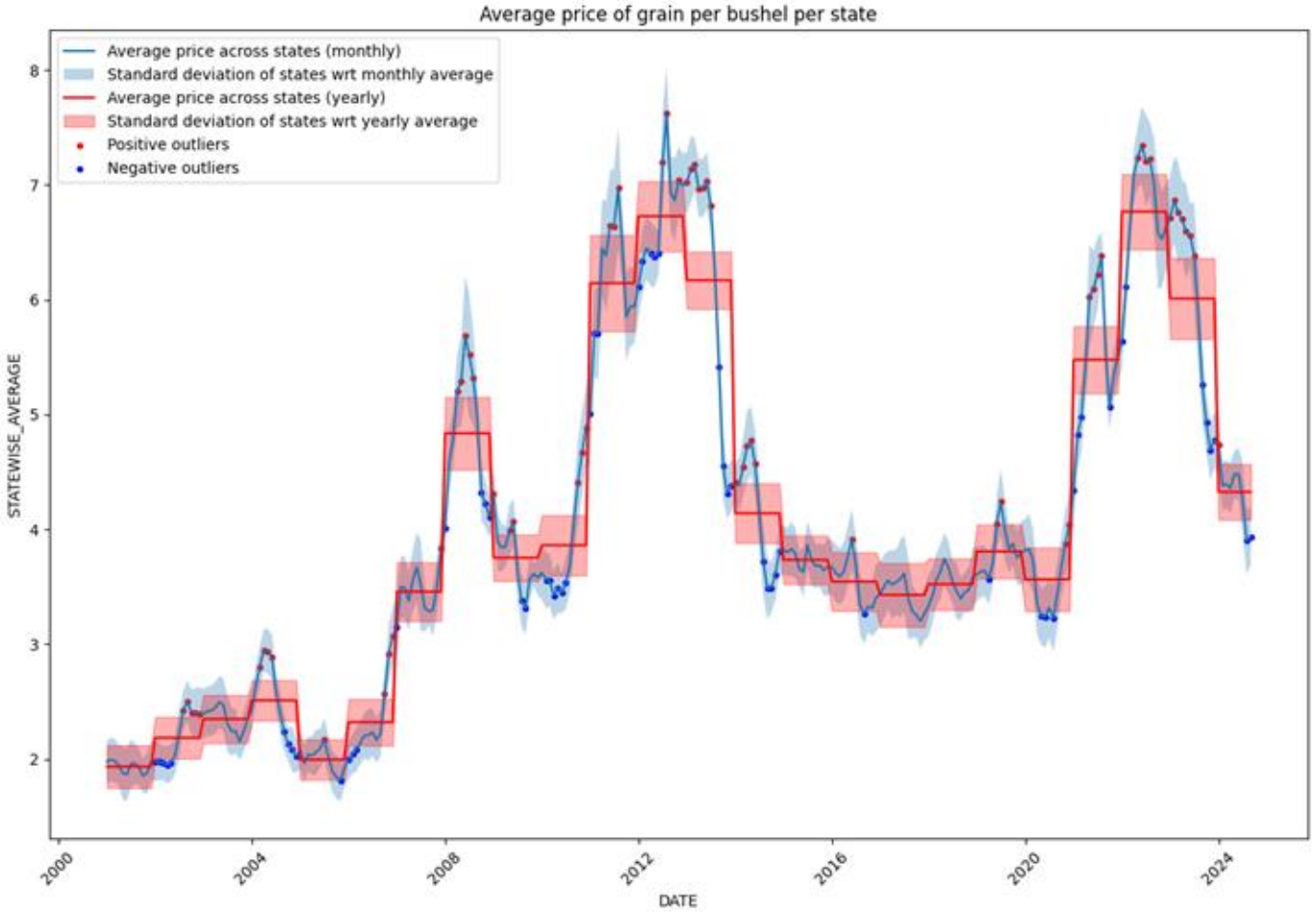
Description

- State - wise representation of monthly corn prices
- 2001 - 2024 time bracket

Observation(s)

- All states behave roughly the same along time
- Prices seem to have generally grown, although not monotonically

MONTHLY CORN PRICE



Description

- State - wise average of corn prices
- 2001 - 2024 time bracket

Observation(s)

- Monthly price outliers highlighted

MONTHLY CORN PRICE

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3

Hypothesis Testing & ML Analysis

Perform statistical hypothesis testing alongside ML analysis to extract knowledge from data

HYPOTHESIS 1

Weather and price outliers delayed correlation

First Step

Do weather outliers affect price ones with a delayed effect?

Do particular weather conditions immediately affect strong variations in price?

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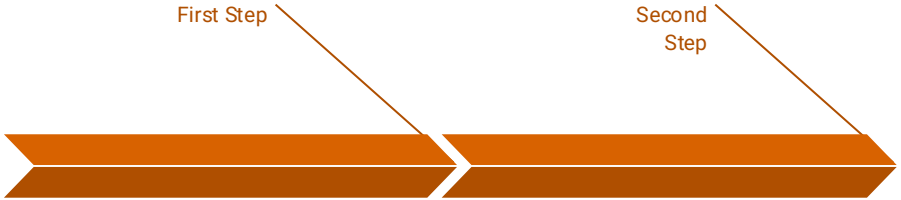

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NULL: Correlation between weather and price outliers is not stronger within the same month than with one month delay

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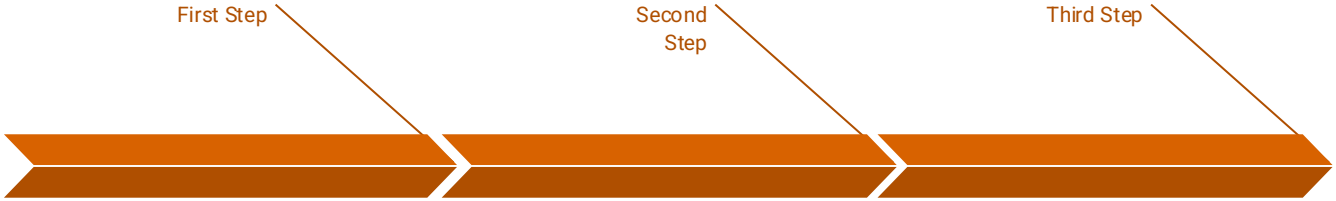

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Third Step

Compute statistic

Compute average correlation among all kinds of weather and price outliers within same month and with one month delay

Fourth step

Evaluate p-value

We were **NOT** able to **REJECT** the null hypothesis with confidence level 5%.

HYPOTHESIS 2

Fluctuation outliers mean

How outliers in the market behave?

```
Reject the null hypothesis with confidence alpha=0.05 with a p-value of 3.014718698804985e-21
T-statistic:  10.226455881986663
Mean difference:  1.18
```

RELATIONSHIP WEATHER - PRICE

Model	**Random forest**
Output	**Corn price**
Features	**Minimum temperature**, **maximum temperature** and **precipitation average** in the corn belt for the previous 12 months + current month.

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YEAR	MONTH	TMIN-0	TMAX-0	PRCP-0	TMIN-1	TMAX-1	PRCP-1	TMIN-2	TMAX-2	...
2002	5	8.110570	20.528783	49.074125	4.931985	17.143079	36.991081	-4.034405	7.600723	...
2002	6	16.577680	28.452915	32.187502	8.110570	20.528783	49.074125	4.931985	17.143079	...
2002	7	19.065747	31.243035	24.859121	16.577680	28.452915	32.187502	8.110570	20.528783	...

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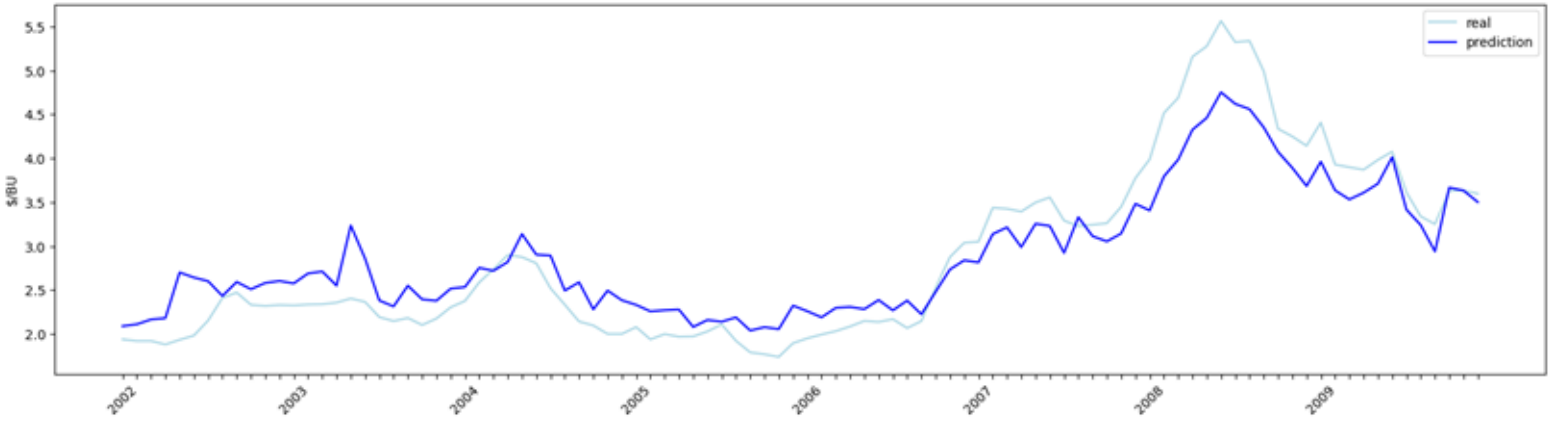
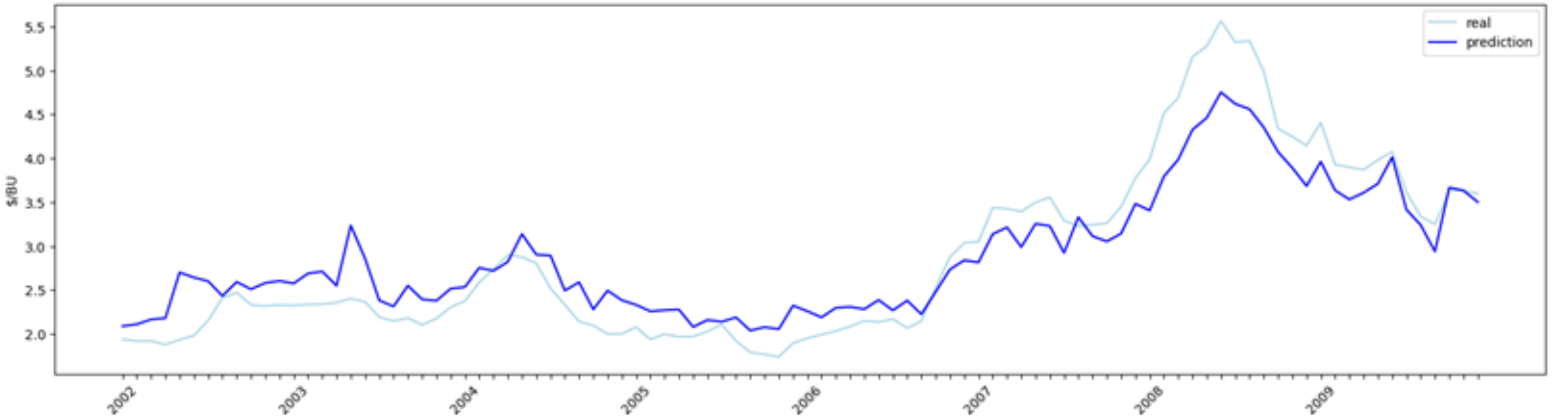
PROBLEM: this approach doesn't provide good results, maybe due to the variable trends in the market, which can be influenced by several factors. For example, during an economic crisis, even if the weather is good and production increases, prices can be high.

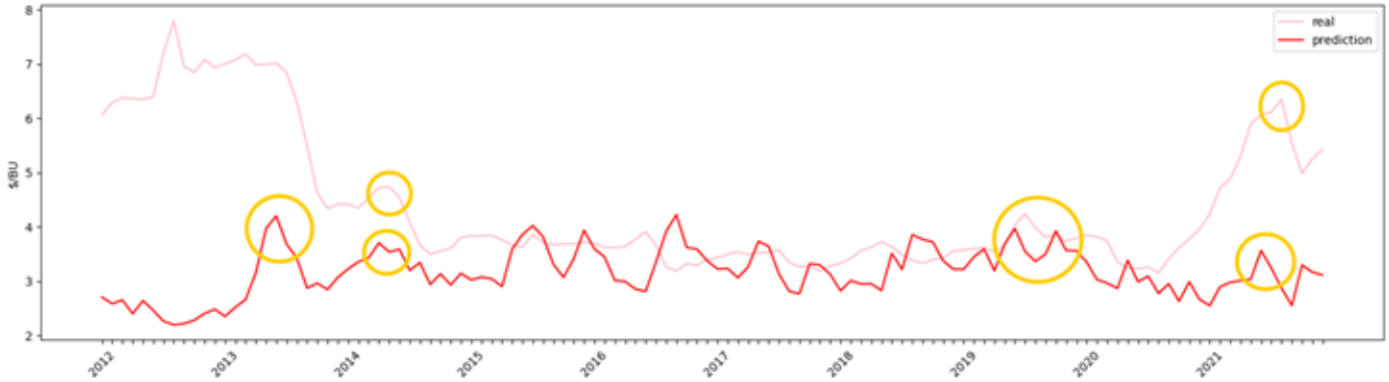
TRAINING


real
prediction
$/BU
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
2002
2003
2004
2005
2006
2007
2008
2009

TRAINING



PREDICTION



MSE = 3.738

MAE = 1.306

RELATIONSHIP OUTLIERS WEATHER - PRICE

Model	**Linear regression**
Output	**Economical outlier**: positive and negative months
Features	**Colder, hotter, drier and wetter month,** that is for months with the highest number of outliers when the seasonal trend is subtracted (from weather data cleaning).

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YEAR	MONTH	colder	hotter	drier	wetter
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1992	7	0	0	0	1
1992	8	1	0	0	0

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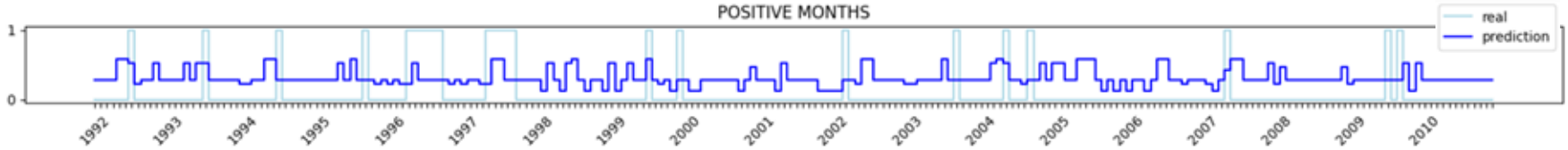
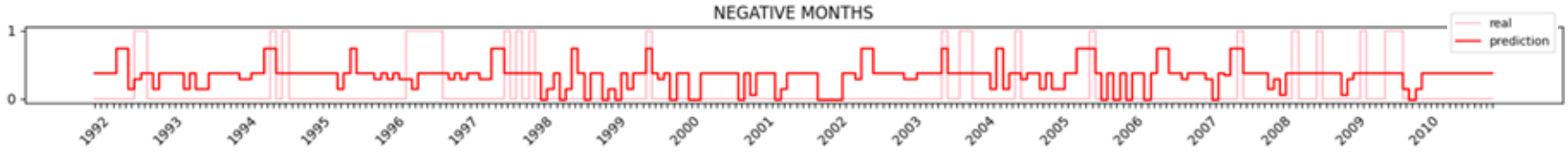
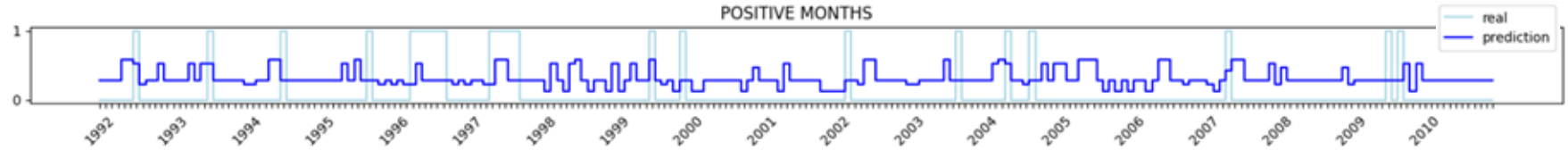
CONCLUSION: a lot of false positives and false negatives:
- very weak correlation between exceptionality of weather and price
- or the employed model is not enough for the considered model.

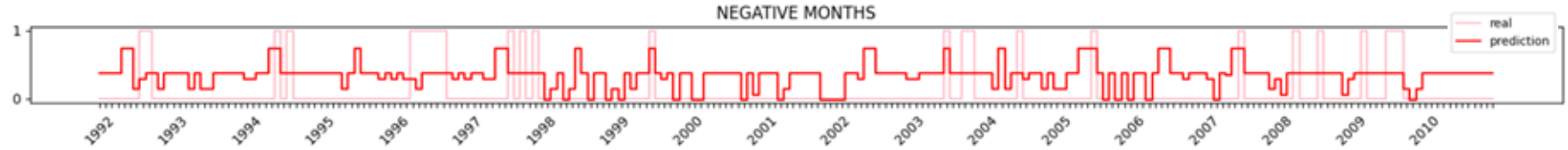
TRAINING


POSITIVE MONTHS
real
prediction
1
0
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010


NEGATIVE MONTHS
real
prediction
1
0
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010

TRAINING





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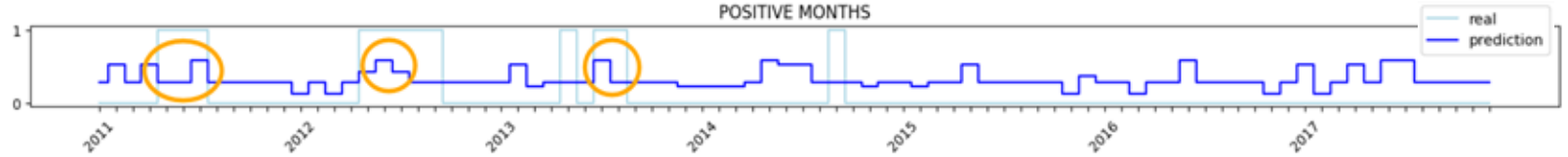
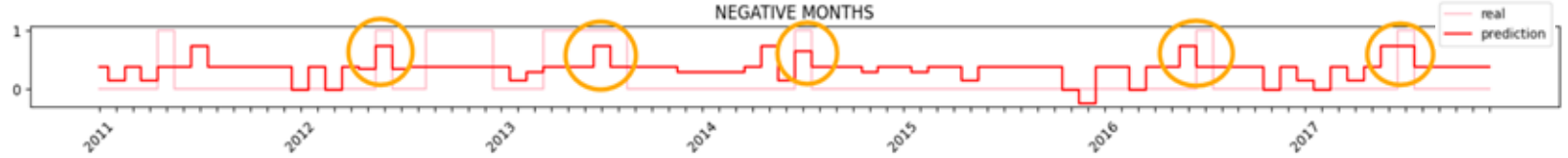
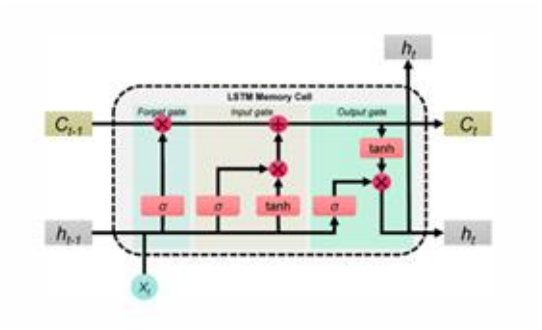
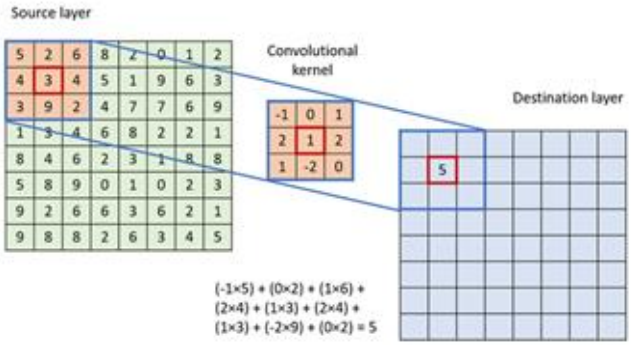


MAE = 0.213

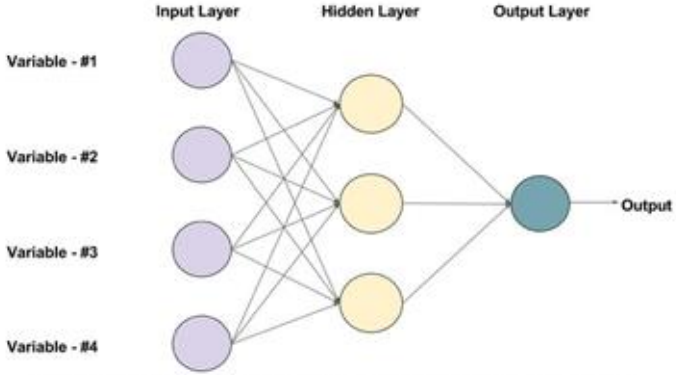


MAE = 0.233

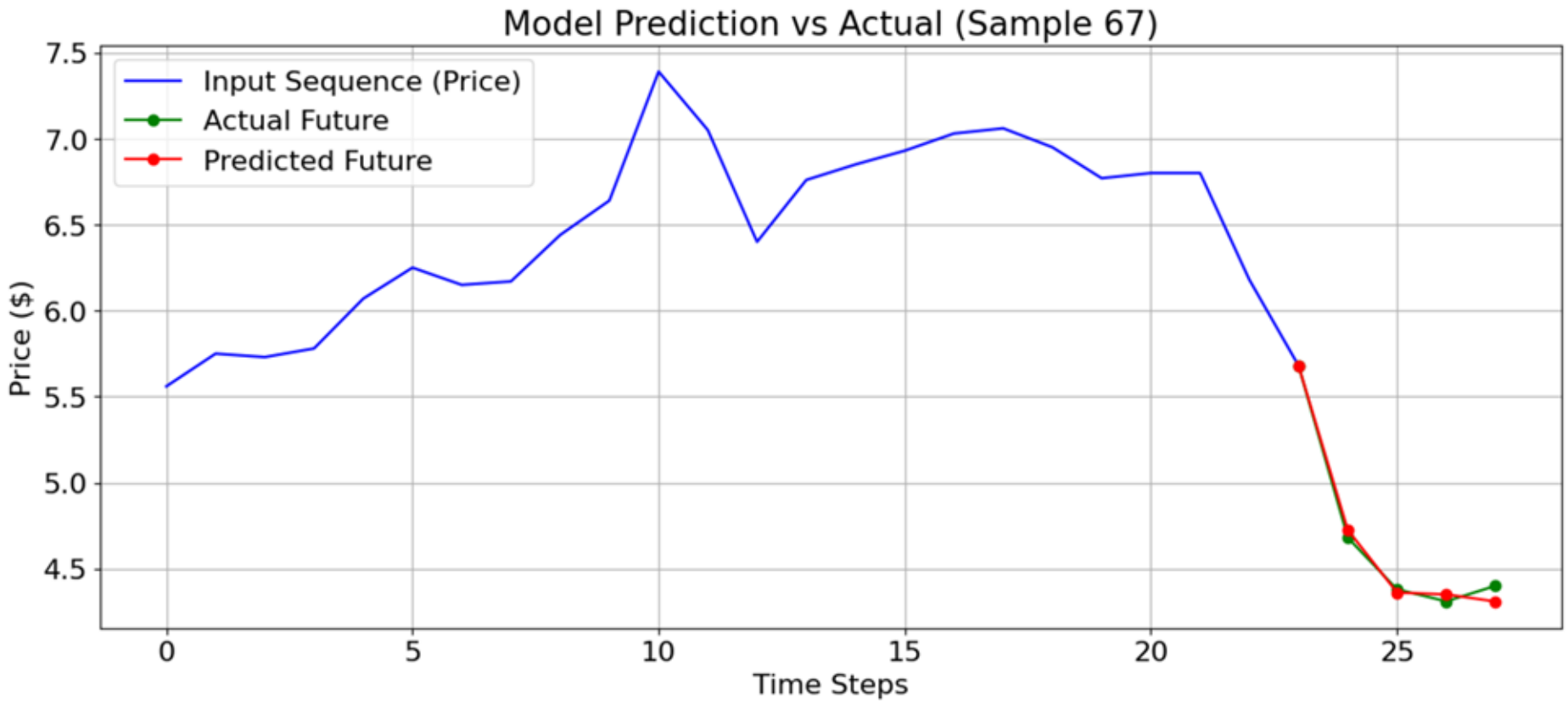


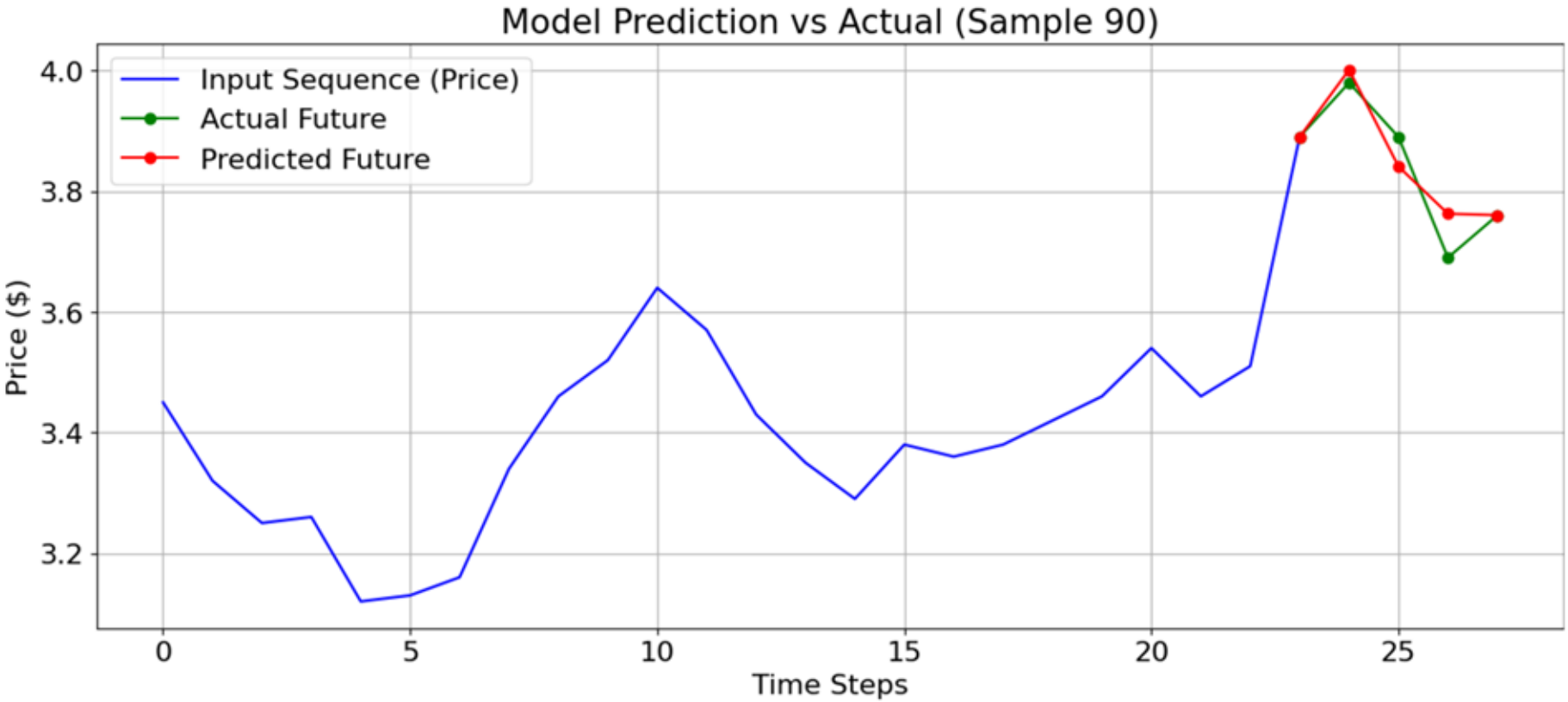


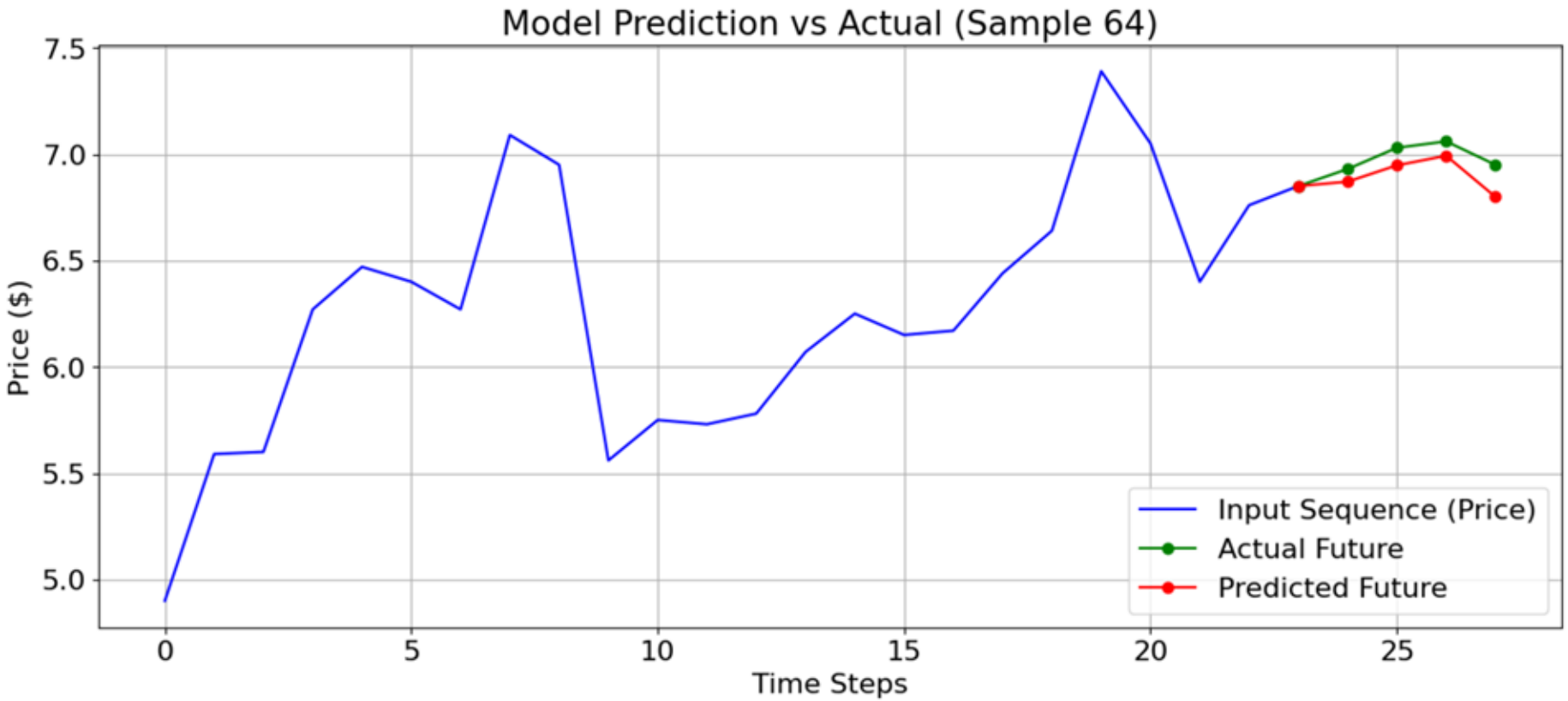
LSTMs + convolution + feed forward (dense)



An example of a Feed-forward Neural Network with one hidden layer (with 3 neurons)


Model Prediction vs Actual (Sample 67)
Input Sequence (Price)
Actual Future
Predicted Future
Price ($)
7.5
7.0
6.5
6.0
5.5
5.0
4.5
0
5
10
15
20
25
Time Steps


Model Prediction vs Actual (Sample 90)
Input Sequence (Price)
Actual Future
Predicted Future
Price ($)
4.0
3.8
3.6
3.4
3.2
0
5
10
15
20
25
Time Steps


Model Prediction vs Actual (Sample 64)
7.5
7.0
6.5
6.0
5.5
5.0
Price ($)
0
5
10
15
20
25
Time Steps
Input Sequence (Price)
Actual Future
Predicted Future

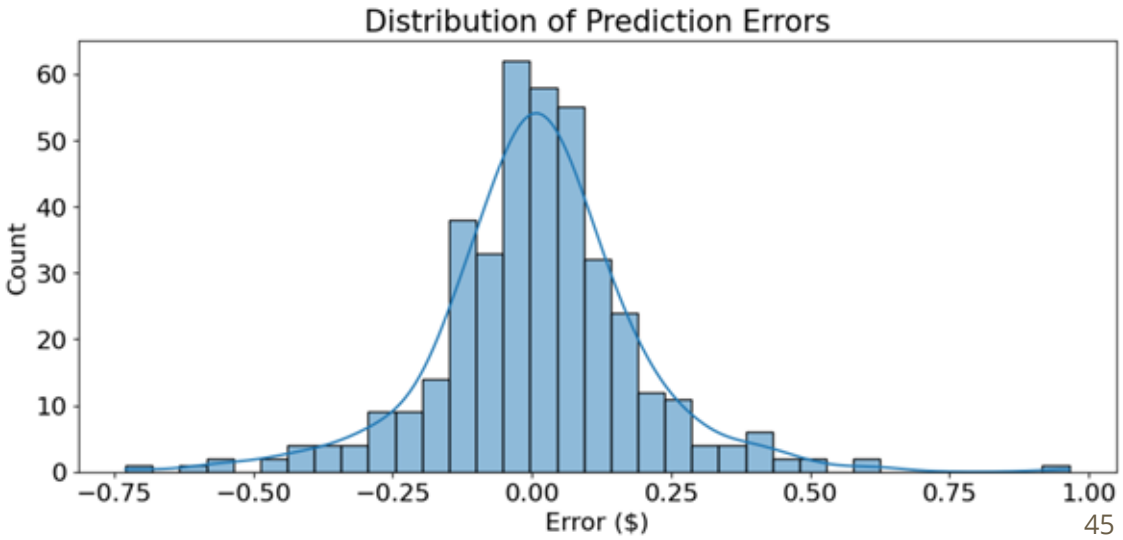
Layer (Type)	Output Shape	Param #
Input (InputLayer)	(None, 24, 4)	0
LSTM	(None, 24, 128)	68,096
Conv1D_3 (Conv1D)	(None, 24, 128)	49,280
MaxPooling1D_2 (MaxPooling1D)	(None, 12, 128)	0
Conv1D_4 (Conv1D)	(None, 12, 256)	98,560
MaxPooling1D_3 (MaxPooling1D)	(None, 6, 256)	0
Conv1D_5 (Conv1D)	(None, 6, 512)	393,728
GlobalAveragePooling1D_1	(None, 512)	0
Dropout_2 (Dropout)	(None, 512)	0
Dense_3 (Dense)	(None, 512)	262,656
Dropout_3 (Dropout)	(None, 512)	0
Dense_4 (Dense)	(None, 128)	65,664
Dense_5 (Dense)	(None, 4)	516

EVALUATION

Model	**Neural Network (custom)**
Output	**Corn price** for the next 4 months
Features	**Minimum temperature**, **maximum temperature** and **precipitation average** in the corn belt for the previous 24 months.

R^2 = 0.98

Directional accuracy = 76.4%



THANK YOU !